

Overview

Build an **AI-powered Data Analyst** that enables users to upload one or more CSV files and interact with the data using natural language.

The application should answer questions, generate insights, create visualizations, detect anomalies, and explain its reasoning using an LLM. The goal is to demonstrate your ability to build a production-ready AI application, not just integrate an LLM API.

Requirements

Your application should support the following:

Core Features

- Upload and validate one or more CSV files
- Answer questions in natural language
- Generate business insights and summaries
- Create charts (Bar, Line, Pie, Scatter, etc.)
- Generate SQL and/or Pandas code when appropriate
- Detect anomalies and explain why they were flagged
- Explain the reasoning behind responses
- Maintain conversation context throughout the session

Example Questions

- Which region generated the highest revenue?
- Show monthly sales trends.
- Which products are underperforming?
- What are the top five customers?
- Generate SQL for this analysis.
- Detect anomalies in the dataset.

Bonus (Optional)

You may implement any additional features such as:

- Multi-file analysis
- Dashboard generation
- Data quality checks
- Forecasting
- Agentic workflows
- Tool calling
- Semantic search
- Caching
- Authentication
- Export reports
- Streaming responses
- Observability/Logging
- Evaluation framework

Technical Expectations

You are free to choose your preferred technology stack.

Your solution should demonstrate:

- Clean, modular code
- Good software engineering practices
- Proper error handling
- Documentation
- Testing (where applicable)

Deliverables

Submit a GitHub repository containing:

- Complete source code
- README with setup instructions
- Architecture diagram
- Both Screenshots and short demo video or Live app link in readme
- Docker support (preferred)
- Sample dataset(s)

Submission

Please submit:

- GitHub repository link
- A **10–30 second demo video** showcasing the application's key features
- Any assumptions or implementation notes in the README